Results

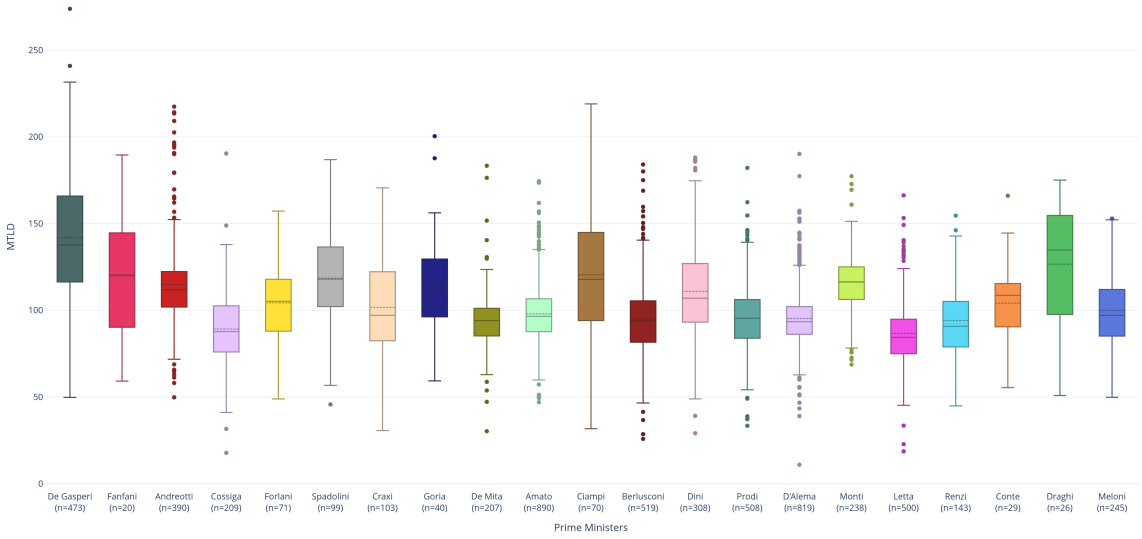
This chapter will present the results from the analysis on the dataset.

Textual Complexity

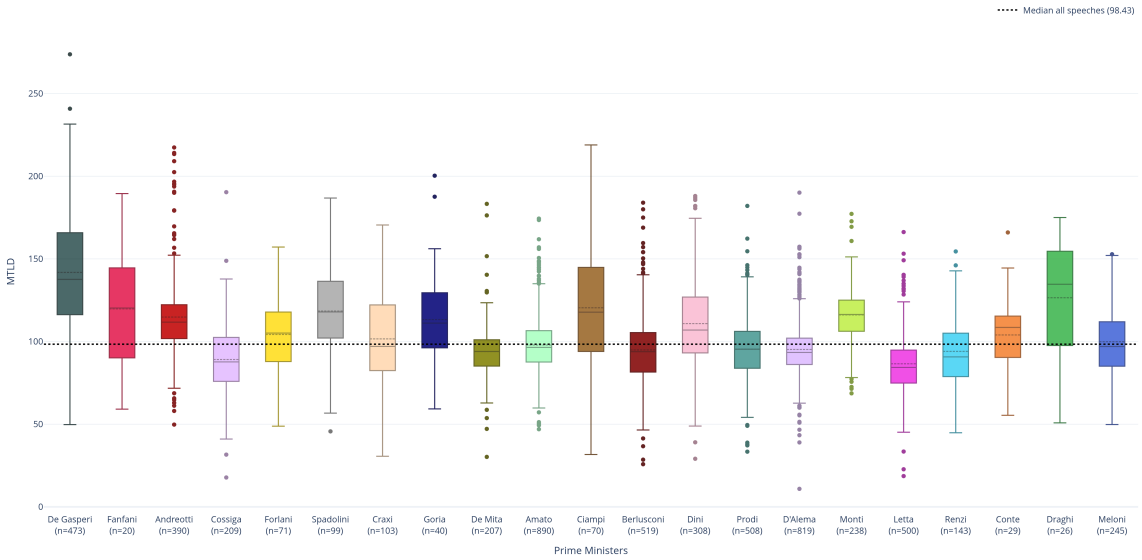
Prime Minister's MTLDs

Let's first take a look at the Prime Ministers themselves. Figure Figure 1 shows a box plot where each box is a Prime Minister with at least twenty speeches, ordered by variability (whisker-to-whisker length) and median diversity respectively. We can visually divide the Prime Ministers into two groups: high- and low-variance. The first group, with seven out of twenty-five Prime Ministers (Ciampi, De Gasperi, Craxi, Fanfani, Spadolini, Dini, and Draghi) exhibit high variability in speeches complexity, suggesting not only a wider vocabulary but also diverse speech styles. This is corroborated by their standard deviations (refer to table Table 1 for the exact values), all above the average of 24.32 (Ciampi, De Gasperi, Fanfani, and Draghi are even above thirty). All the other Prime Ministers, pertaining to the low-variance group, seem more consistent in the complexity of their speeches, and their standard deviations are all under the average. That being said, between Forlani and De Mita the inter quartile range halves, going from 28.76 to 15.86 for a reduction of 44.85%, telling us that even in the low-variance group there are strong differences in variance between Prime Ministers. In this last group, two exceptions emerge: Conte, whoose standard deviation of 24.32 sits exactly on the median, and Goria with a standard deviation of 29.26. A possibile explanation could be found in their low speech count (twenty-nine and fourty respectively) which would give more weight to their high-complexity outliers. Other Prime Ministers present more outliers (for example D'Alema has fifty-two, Andreotti twenty-eight) which can clearly be seen in the plot, but their standard deviations do not get impacted as much because of their great amount of speeches (819 for D'Alema, 390 for Andreotti) stabilizing the deviation.

Overall, Prime Ministers in this study exhibit variation between one another, suggesting that there is not a "complexity typical of a Prime Minister", and that each one have their own style and vocabulary richness.



(a) Boxplot of MTLD values for each Italian Prime Minster, without overall median.



(b) Boxplot of MTLD values for each Italian Prime Minster, with overall median.

Figure 1: Figure 1a and Figure 1b show MTLD values for each Italian Prime Minister, ordered by their first time in office, allowing for not a direct comparison between Prime Ministers. The mean is a dotted line. Since all speeches for each Prime Minister were collected (outside their time in office) the graph can **not** be interpreted as the evolution of MTLD over time. Prime Minister with fewer than 20 speeches are omitted.

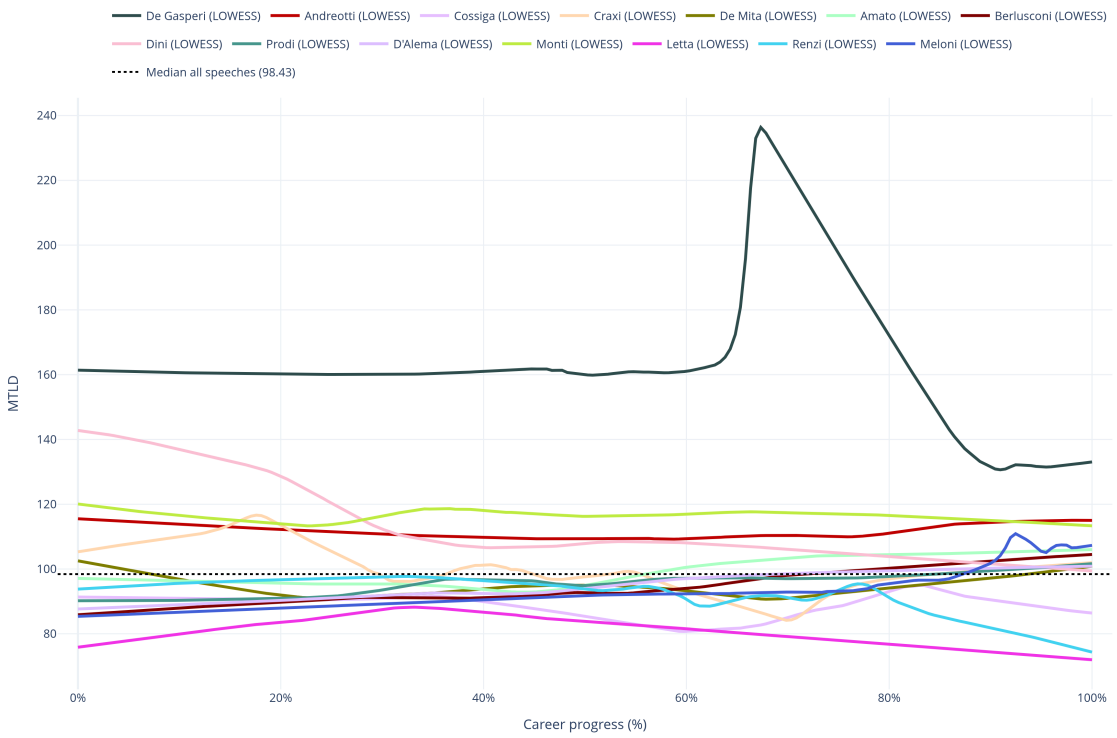
Table Table 1 also shows the slope for each Prime Minister. This represents whether, during their careers, the complexity of their speeches increased, decreased, or remained the same. Most Prime Minister's show no significant trend, with six exceptions: Draghi (+0.462), Conte (+0.384), De Gasperi (−0.330), Meloni (+0.263), Craxi (−0.265), and Dini (−0.236). The trends can also be visualized in figure Figure 2, were a LOWESS regression[1] has been plotted for each Prime Minister's career. Once again, De Gasperi shows as a clear outlier, with a median complexity consistently above the

[1]LOWESS was chosen over linear models or Splines because it combines much of the simplicity of linear least squares regression with the flexibility of nonlinear regression. In short, it is flexible but interpretable.

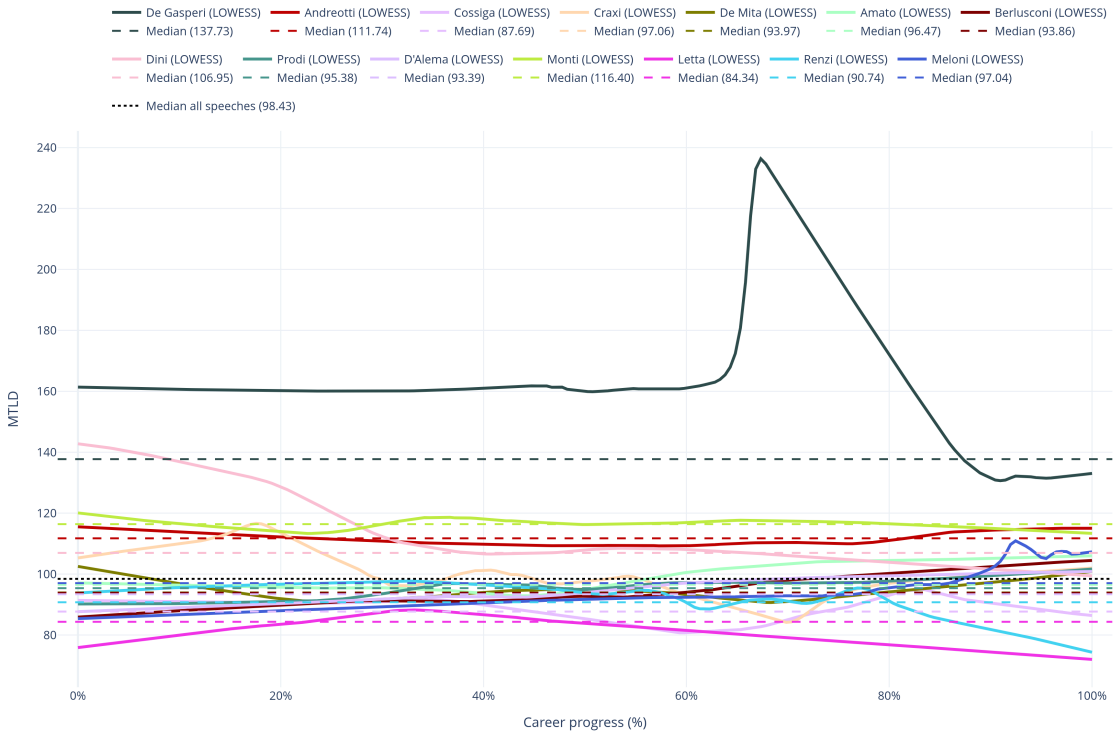
rest, while the other Prime Ministers cluster around the overall median. Draghi, Conte, and Fanfani regressions look less consistent, which is expected since they do not have a lot of data points.

PM	# Speeches	Mean	Median	Std. dev.	IQR	# Outliers	Trend slope (per %)
De Gasperi	473	141.83	137.64	35.88	49.50	2	−0.330
Fanfani	20	119.80	120.31	37.59	50.09	0	−0.130
Andreotti	390	114.81	111.74	23.77	20.42	28	0.025
Cossiga	209	89.12	87.69	21.03	26.41	4	0.180
Forlani	71	104.28	105.13	23.38	28.76	0	0.016
Spadolini	99	118.43	117.94	25.58	34.01	1	0.035
Craxi	103	101.63	97.06	27.29	39.60	0	−0.265
Goria	40	113.30	111.16	29.26	31.68	2	0.120
De Mita	207	93.97	93.97	16.62	15.86	10	−0.060
Amato	890	97.90	96.47	16.07	18.98	25	0.144
Ciampi	70	120.44	117.78	37.15	50.30	0	−0.066
Berlusconi	519	94.92	93.87	21.17	23.83	19	0.194
Dini	308	110.88	106.95	25.97	33.71	8	−0.236
Prodi	508	95.42	95.38	19.81	22.25	16	0.044
D'Alema	819	95.23	93.39	16.08	15.93	52	0.153
Monti	238	116.15	116.40	16.42	18.57	11	−0.034
Letta	500	86.59	84.34	17.37	19.87	18	−0.104
Renzi	143	94.06	90.74	21.15	26.05	2	−0.166
Conte	29	104.02	108.58	24.32	24.67	1	0.384
Draghi	26	126.53	134.68	33.52	55.17	0	0.462
Meloni	245	99.89	97.04	21.20	26.85	1	0.263

Table 1: Descriptive statistics of the MTLD (Measure of Textual Lexical Diversity) per prime minister, ordered by first term in office. In order: Prime Minister's name, number of speeches, mean, median, standard deviation, interquartile range (IQR), number of outliers (speeches outside the whiskers of the boxplot), and slope of the MTLD trend per percentage point. Prime ministers with too few usable speeches are not included.



(a) LOWESS curve of MTLD values for each Italian Prime Minster, without overall median.

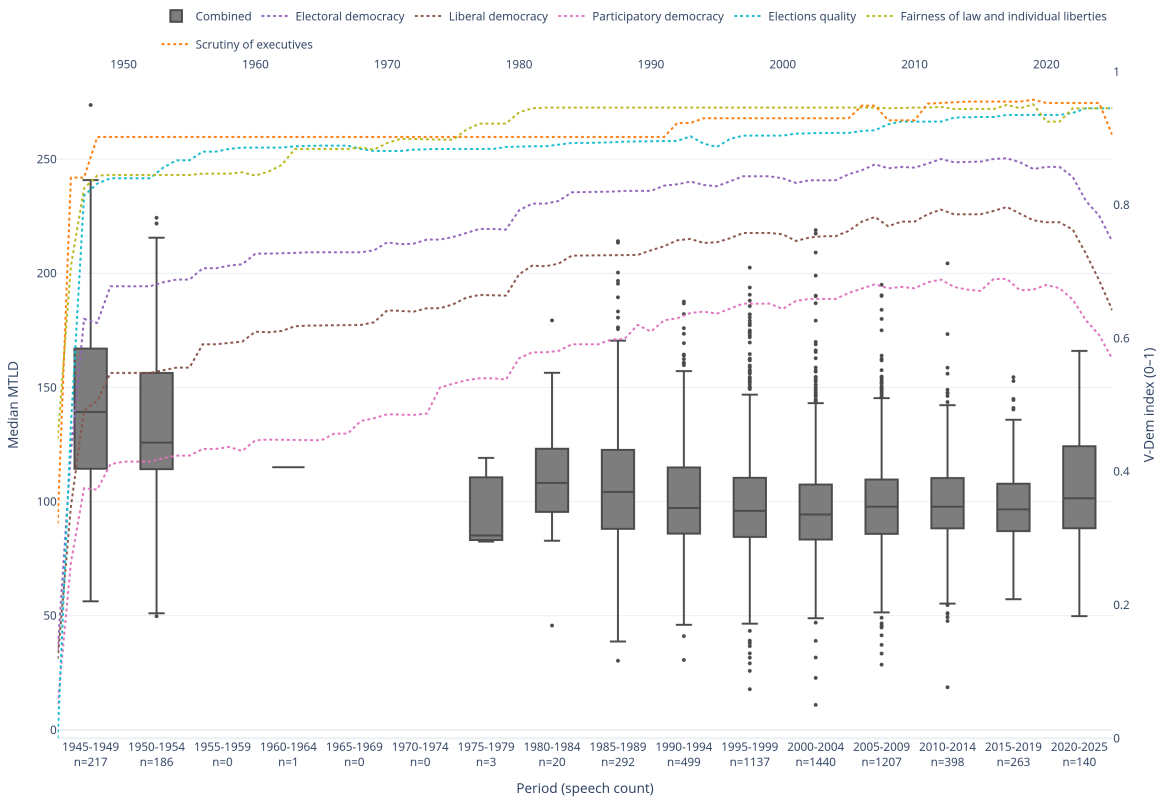


(b) LOWESS curve of MTLD values for each Italian Prime Minster, with overall median.

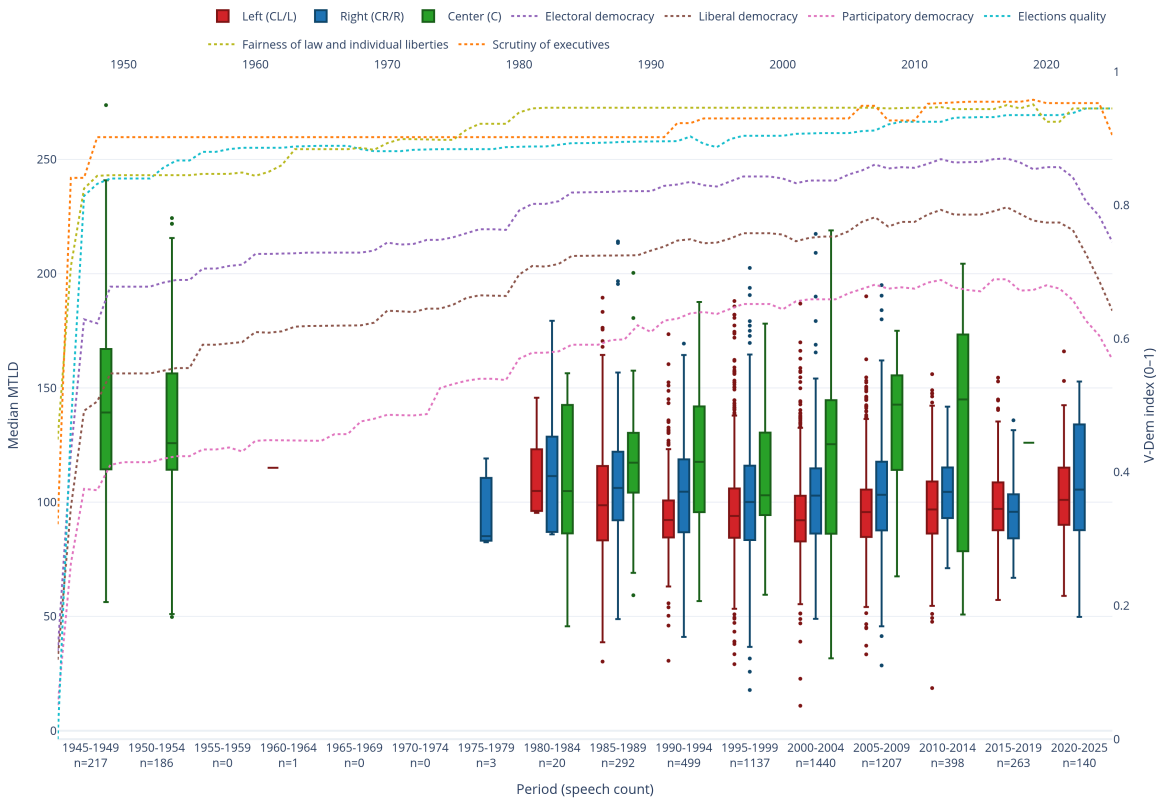
Figure 2: Figure 2a and Figure 2b show MTLD values for each Italian Prime Minster over their careers span. The LOWESS curve allows was chosen to aid in the interpretation of the values. Prime Minister with fewer than 20 speeches are omitted.

VDEM

5 years



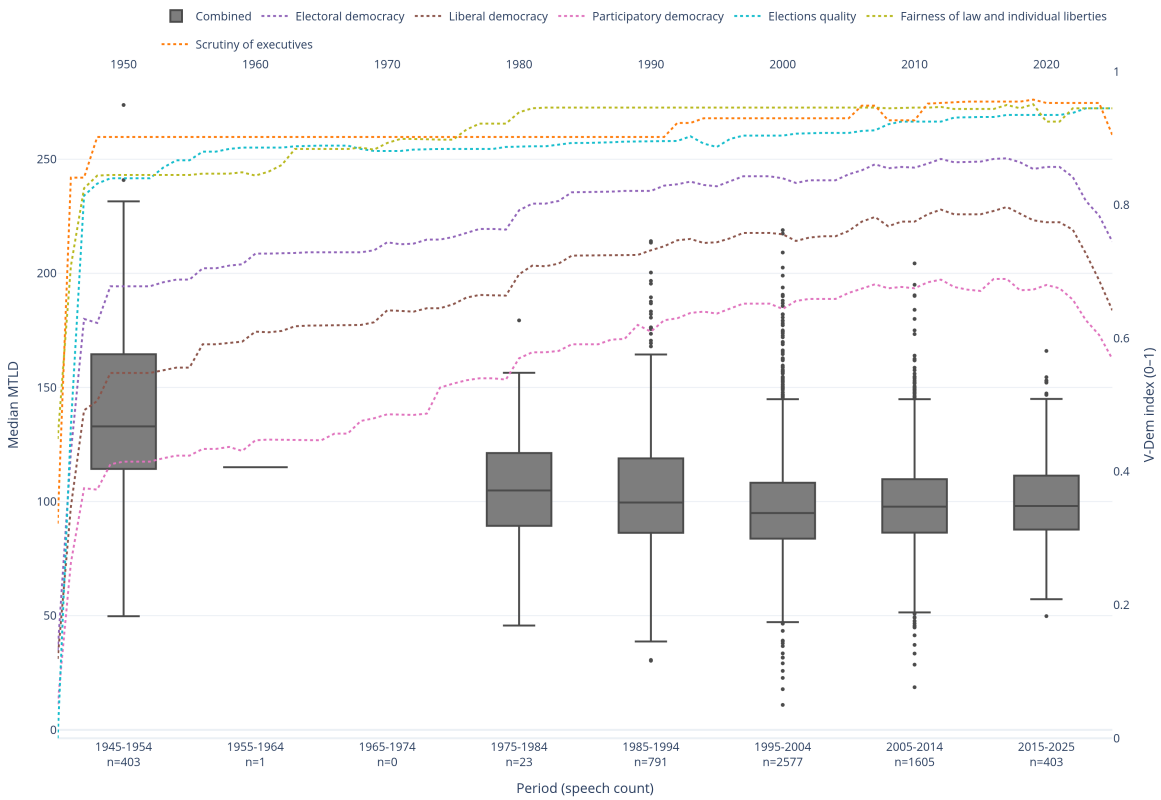
(a) MTLD values in a Boxplot by 5-year periods, with VDEM indices. Each grey box combines all



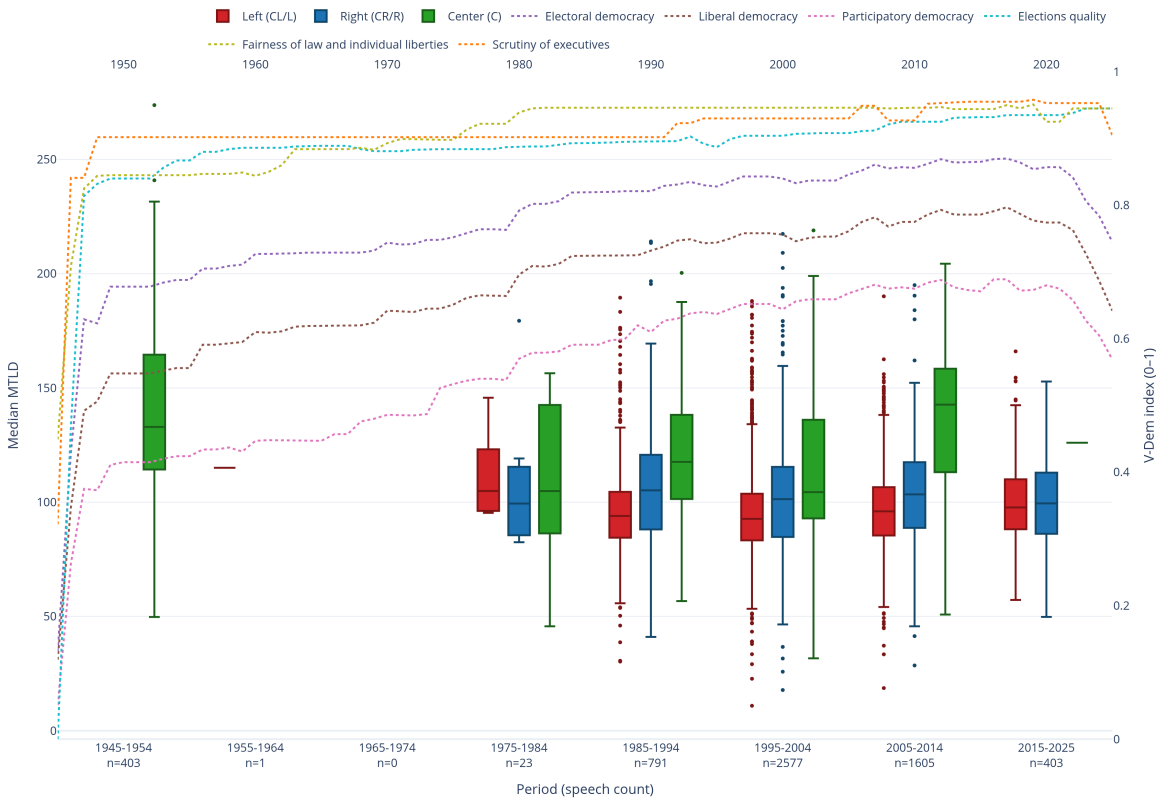
(b) MTLD values in a Boxplot by 5-year periods, with VDEM indices. As above, each grey box combines all speeches for all political leanings in that 5-year period, while the colored boxes represent respectively center-leaning (in green), left-leaning (in red), and right-leaning (in blue) MTLD values in that 5-year preiod.

Figure 3: MTLD values in a Boxplot by 5-year periods. There seem to be a reduction in MTLD variance starting from 1989 all the way to 2019, whith an abrupt inversion in the 2020-2025 period. If this is a correlation with the inflection of some VDEM indices, it is not clear.

10 years



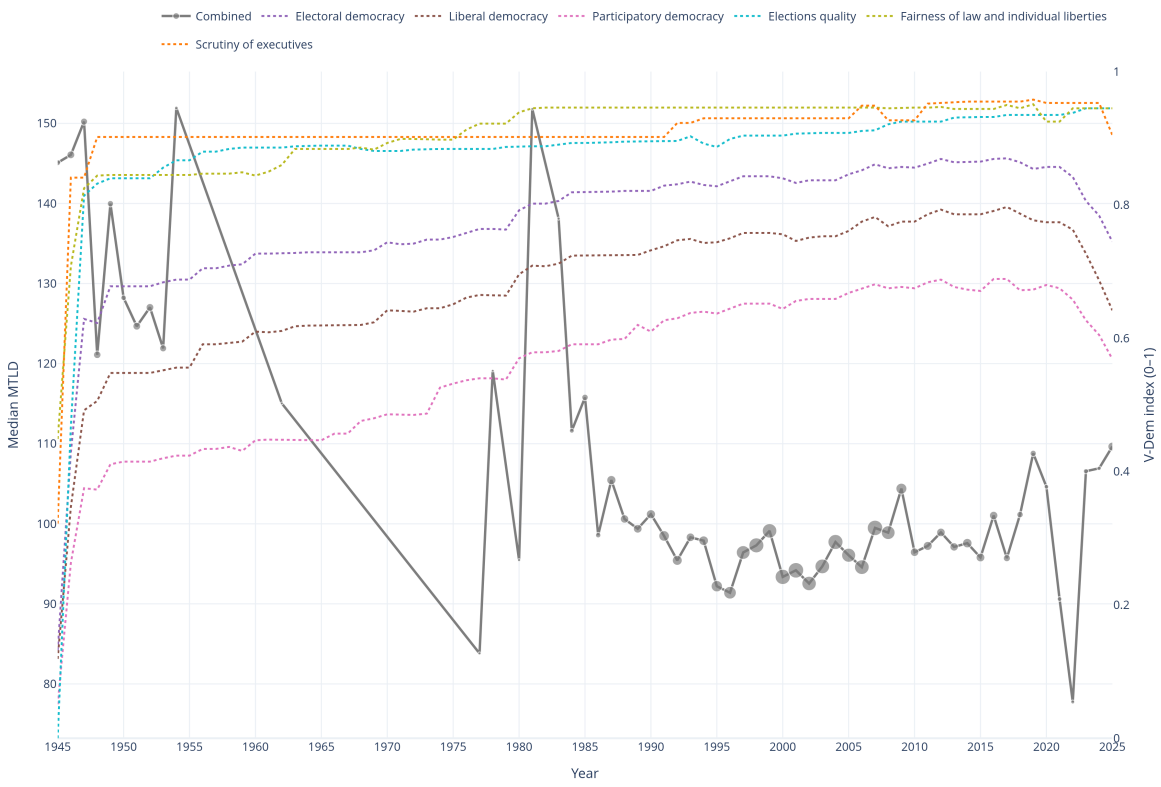
(a) MTLD values in a Boxplot by 10-year periods, with VDEM indices. Each grey box combines all



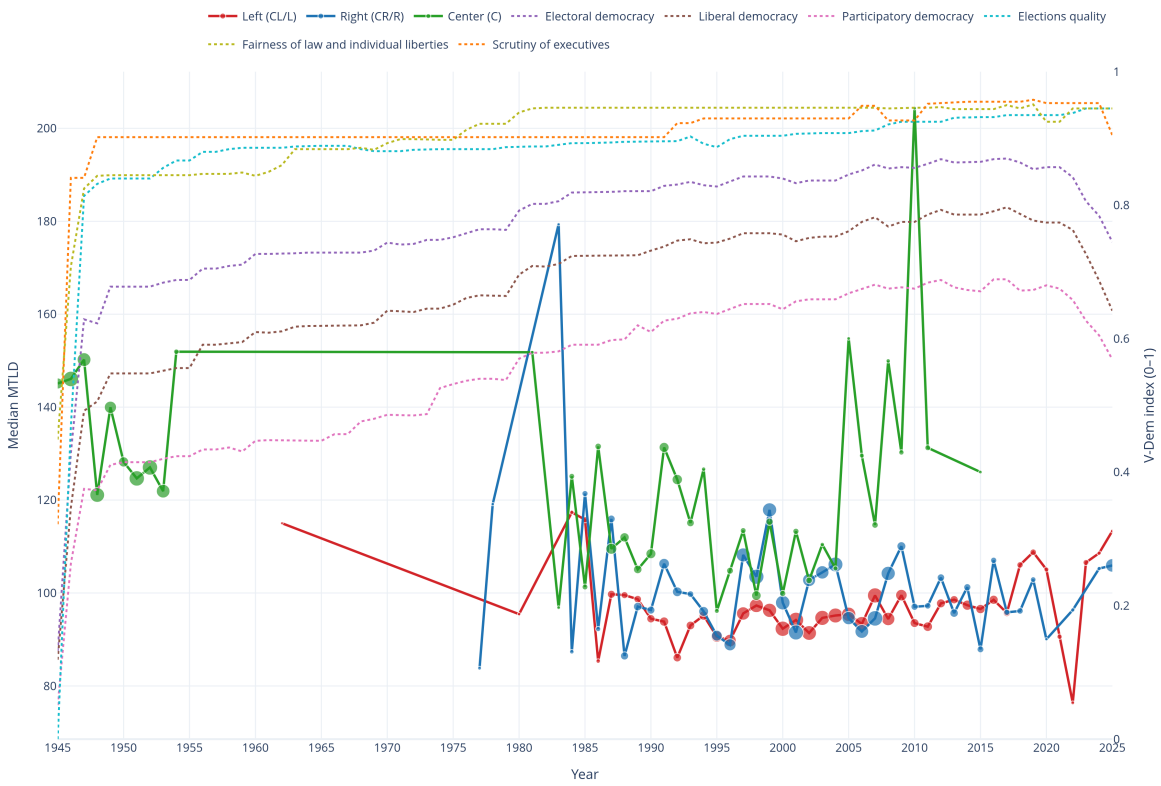
(b) MTLD values in a Boxplot by 10-year periods, with VDEM indices. As above, each grey box combines all speeches for all political leanings in that 10-year period, while the colored boxes represent respectively center-leaning (in green), left-leaning (in red), and right-leaning (in blue) MTLD values in that 10-year preiod.

Figure 4: MTLD values in a Boxplot by 10-year periods. There seem to be a reduction in MTLD variance starting from 1989 all the way to 2025. If this is a correlation with the inflection of some VDEM indices, it is not clear.

All



(a) MTLD values plotted for each year, aggregating all speeches from all political leanings. Each dot



(b) MTLD values plotted for each year, aggregating speeches from three political leanings, respectively colored as green (center), red (left), and blue (right). Each dot of the graph (for each year) is scaled based on the amount of speeches available for that specific year. The MTLD line chart is plotted against VDEM indices.

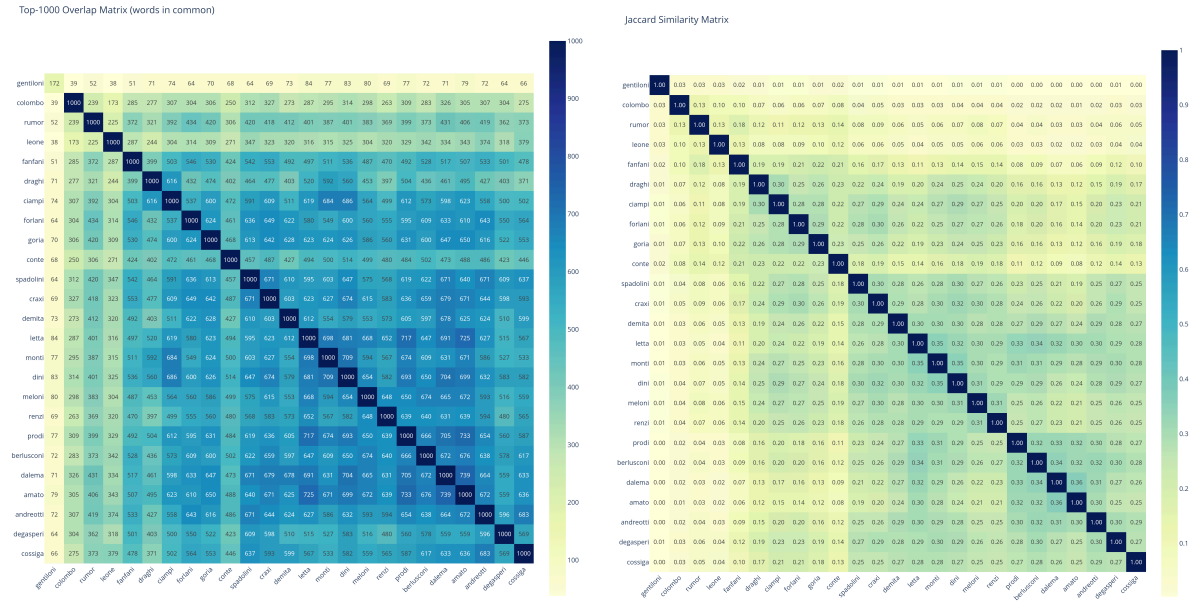
Figure 5: MTLD values aggregated for each year, plotted against VDEM indices. There does not seem to be a correlation between a decrease in complexity and a decrease in democratic indices. Instead, the graph seems to suggest the opposite.

Conclusions

There does not seem to be a clear correlation between MTLD values and democratic indices, with the exception for the variance which ...

Polarization

Prime Minister's Lexical Similarity

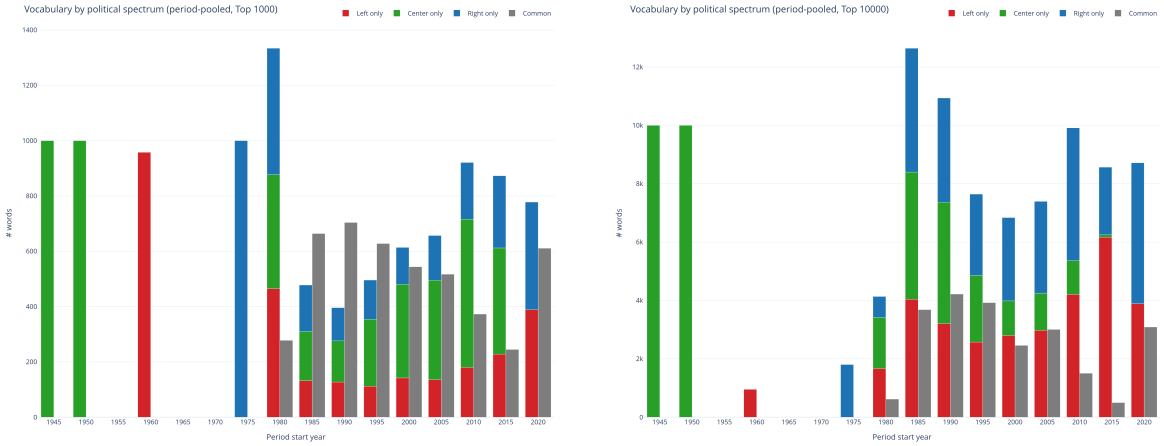


(a) Words similarity.

(b) Jaccard similarity.

Figure 6: Heatmaps showing lexical similarity between Prime Ministers' speeches. Figure 6a was made by pooling the 1000 most used words by each Prime Minister, then calculating how many words are shared between each of them. Figure 6b, on the other hand, is made by calculating the Jaccard index between each pair of Prime Ministers, by pooling *all* words from all speeches used by each Prime Minister.

Political Leanings Lexicons Over Time



(a) Top 1,000 pooling.

(b) Top 10,000 pooling

Figure 7: All speeches are pooled in 5-year time frames. Then, for each 5-year time frame, the top-N most frequently used words are extracted for each political leaning[2] (one thousands words for graph at Figure 7a, and ten thousands words for graph at Figure 7b). This results in three distinct sets[3] of words for left, right, and center leaning speeches respectively.

The graph shows, for each 5-year time frame, words in common among all leanings (in grey), and words unique for each leaning (in red, blue, and green respectively). For the years with most data (from 1985 to 2015), Figure 7a shows a progressive increase in unique vocabularies compared to a decrease in shared ones, which can be interpreted as symptoms of increasing polarization.

[2] A speech is considered "right-leaning" if it pertains a right-leaning Prime Minister.

[3] When speeches for each leaning are present. Otherwise only two, one, or no set at all can be extracted from that 5-year time frame.

t-SNE

Word2Vec: Word Embeddings

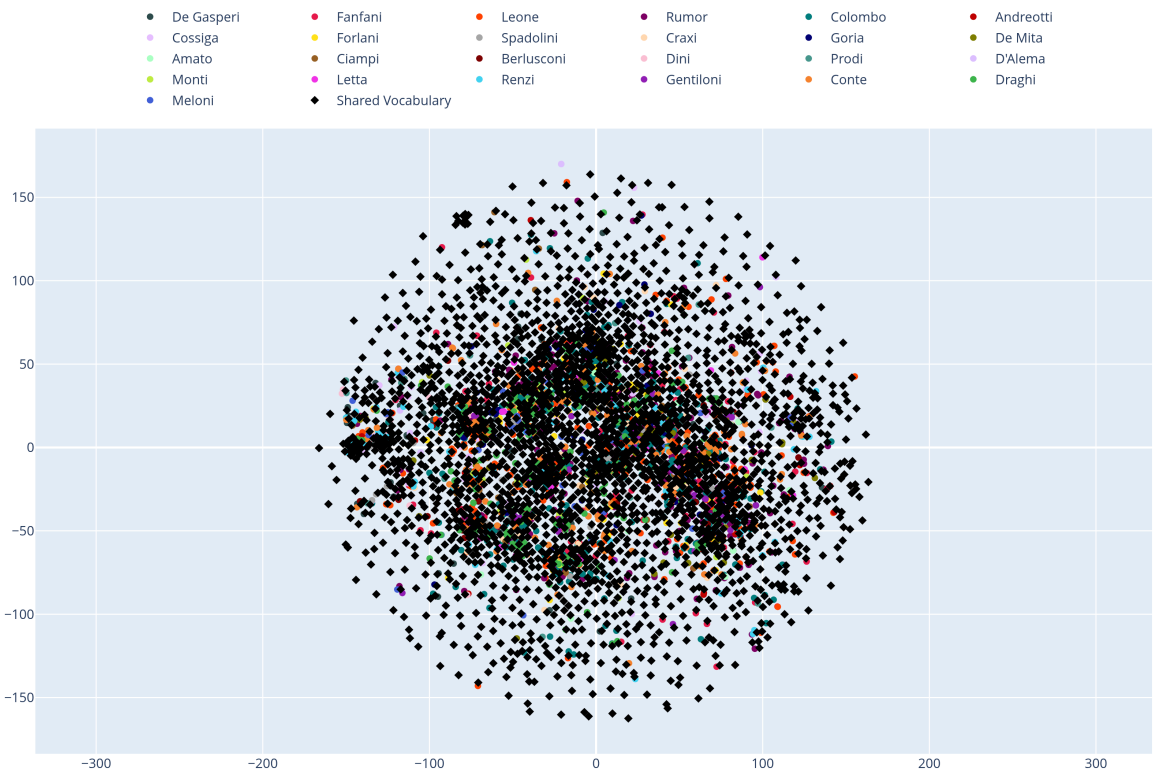
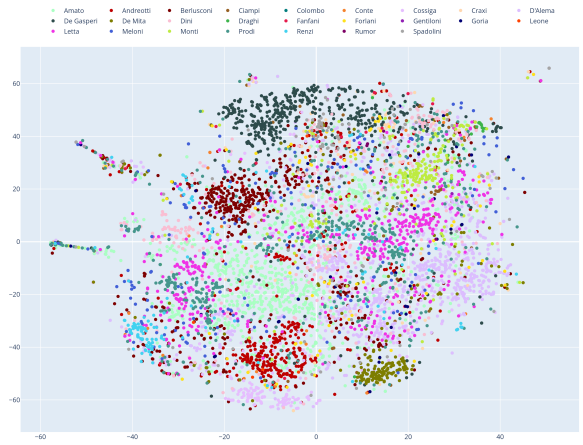


Figure 8: Top-1000 words[4] from each Prime Minster. The embedding vectors are pretrained Italian word vectors (CBOW, 300 dimensions) taken from *fastText*. Words shared by all Prime Ministers are displayed as black rhombuses. The graph show a lot of clustering, suggesting a certain similarity among Prime Ministers. That being said, the graph is also too hard to read to gain any conclusive evidence.

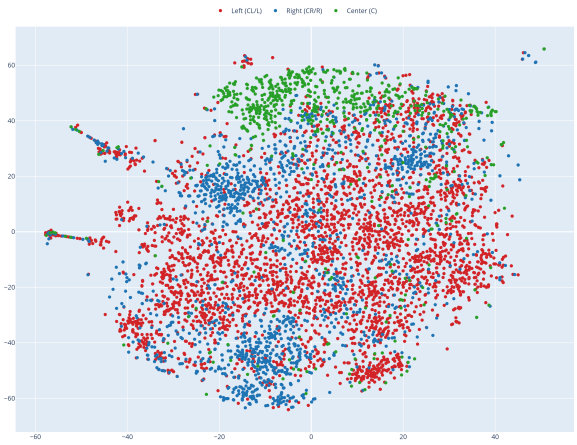
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[4]Top-1000 words means pooling the one thousand most used words of a specific Prime Minister. Stopwords are removed and the lexicon is lemmatized.

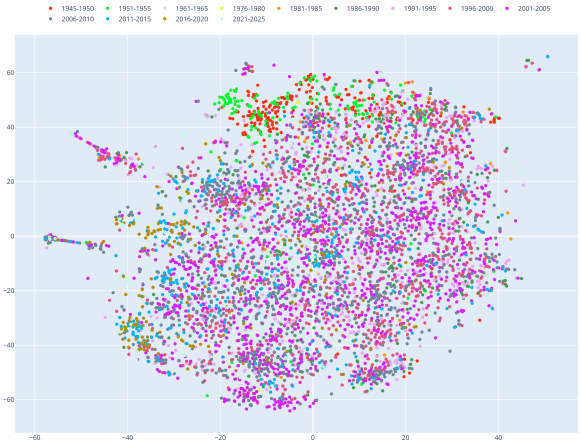
Doc2Vec: Speech Embeddings



(a) Doc2Vec: Prime Ministers



(b) Doc2Vec: political leaning



(c) Doc2Vec: year periods

Figure 9: Using `TfidfVectorizer` from `sklearn` library, I computed one embedding for each speech[5]. The embeddings can then be projected in two dimensions. Figure 9a shows the projected speech vectors grouped by Prime Minsters. We can see clustering, suggesting that most prime ministers have a lexical style and wording that are unique to them. Figure 9b groups the vectors by political leaning, showing that different leanings occupy different and distinct regions in the vector space. Both graphs strongly hints to a polarized political landscape.

On the other hand, Figure 9c shows two main clustering regions: on the top, speeches from 1945 to 1955 gather, while speeches from 1991 to 2025 are more uniformly scattered across the whole vector space. This suggests that earlier speeches were significantly different compared to modern ones, while speeches from the nineties onward are quite similar.

TODO: compare with VDEM polarization indices (if exists).

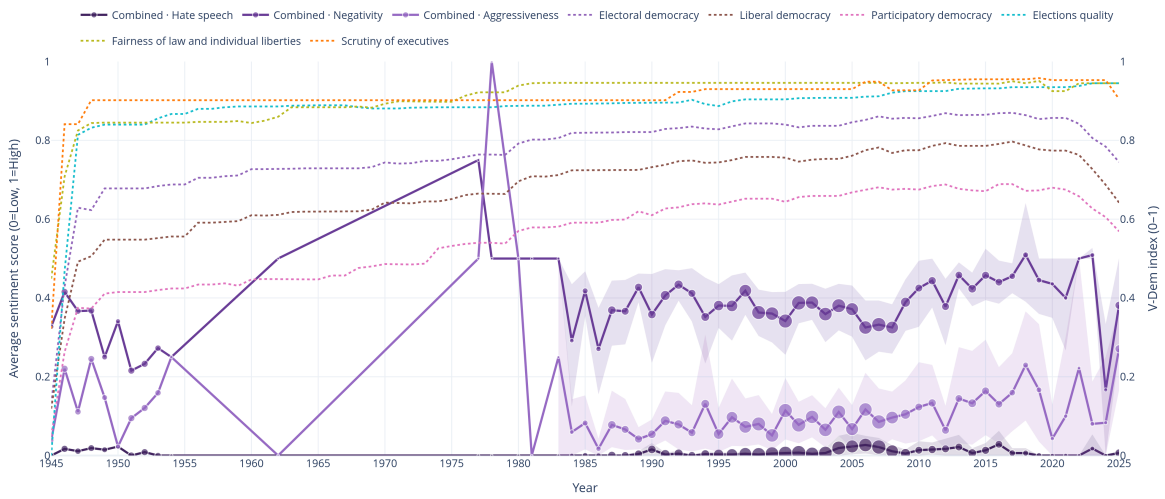
Conclusions

We see quite a bit of polarization, but ...

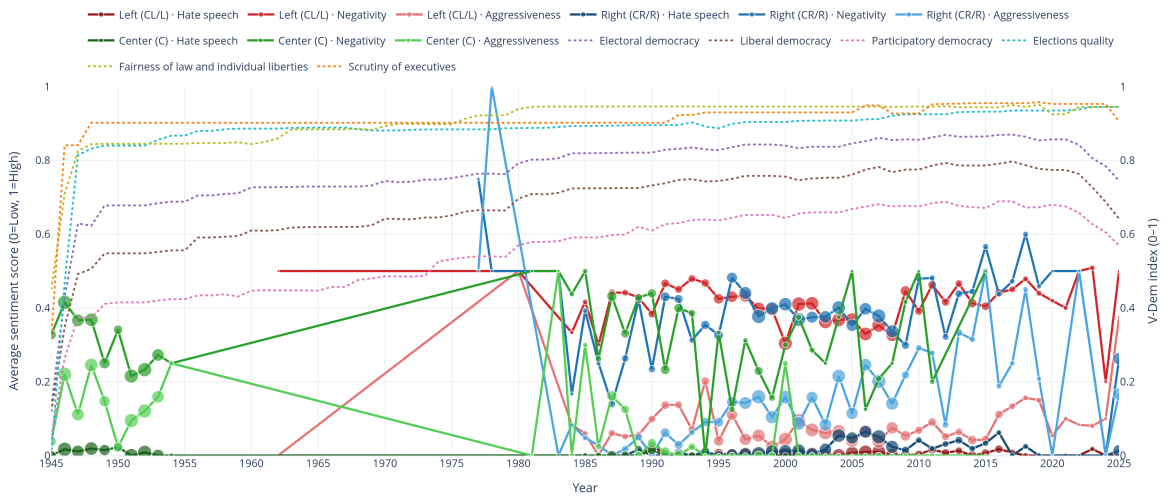
[5]Each speech counts as a document.

Sentiment Analysis

Democratic Indices Against Sentiment Scoring



(a) Sentiment values plotted for each year, aggregating all speeches from all political leanings. Each dot of the graph (for each year) is scaled based on the amount of speeches available for that specific year. The confidence intervals are shown. The sentiment scores chart is plotted against VDEM indices.

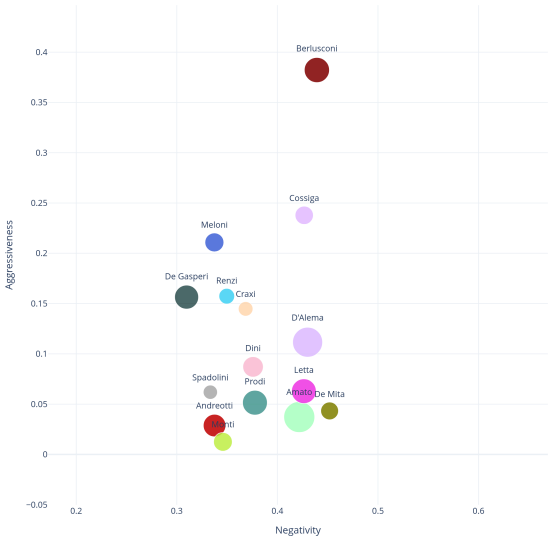


(b) Sentiment values plotted for each year, aggregating speeches from three political leanings, respectively colored as green (center), red (left), and blue (right). Each dot of the graph (for each year) is scaled based on the amount of speeches available for that specific year. The The sentiment scores line chart is plotted against VDEM indices.

Figure 10: Sentiment analysis results for *hate speech*, *aggressiveness*, and *negativity* plotted against VDEM indices. Its hart to tell wether there is any correlation between them.

Toxicity (TODO: change name)

First thing first: hate speech is so low for everyone that we can ignore it



(a) Sentiment: Prime Ministers ($|\text{speeches}| \geq 100$).



(b) Sentiment: leanings.



(c) Sentiment: years zoomed out.



(d) Sentiment: years zoomed in.

Figure 11: Sentiment analysis results for *negativity* and *aggressiveness*. Values go from zero to one, and the classification task considers three possibile values for each speech: *low* (0), *mid* (0.5), and *high* (1).

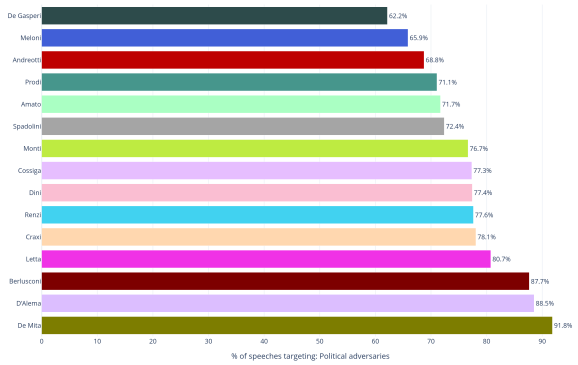
Size of bubbles depends on the amount of speeches. Figure 11a project results for single Prime Ministers: Berlusconi stands out as the most aggressive by far. Figure 11b project results aggregated by leanings. The right appears to be more aggressive, while the left more negative. The center seems to be more positive than both. The last two figures show the sentiment scores aggregated by years (with 5-year pool range). While Figure 11c shows an outlier for the period 1976 — 1980, Figure 11d shows that overall the sentiment scores are quite close to each other, without great variance especially for periods with a lot of speeches.

Targets

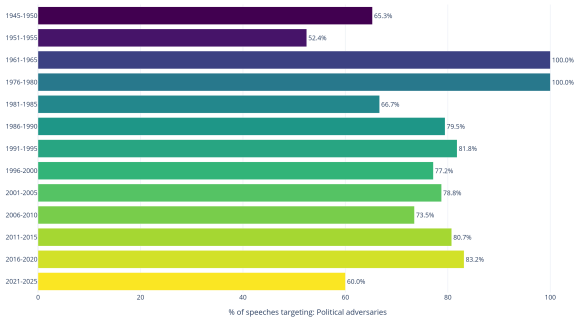
Five possible targets:

- None;
- Political adversaries;
- Gender minorities;
- Religious minorities;
- Ethnic minorities.

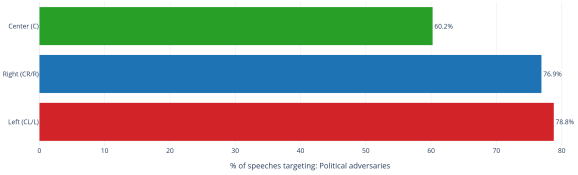
The targets are classified independently to the sentiment scores. For example, a speech could have *low* aggressiveness and still be targeting political adversaries.



(a) Target: political adversaries. Grouped by: PMs.



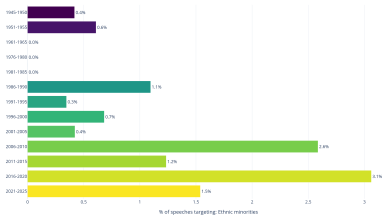
(b) Target: political adversaries. Grouped by: years.



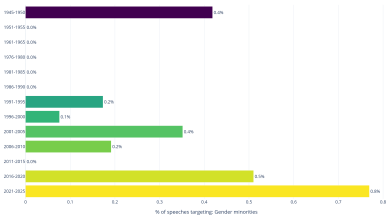
(c) Target: political adversaries. Grouped by: leanings.

Figure 12: All graphs show the percentage of speeches that target political adversaries. Figure 12a shows which Prime Minister speaks more often about the opposition (prime ministers with less than 100 speeches are omitted). Figure 12b shows in which 5-year periods there are more speeches targeting political adversaries (ordered by year). Lastly, Figure 12c shows which of the three political fields is more likely to speak about the opposing front. The results show how, across the board, more than one speech out of two targets the opposition. This is true regardless of time period or affiliation.

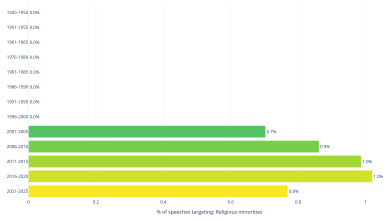
Thankfully, Italian Prime Minister apparently do not target minorities often.



(a) Target: ethnic minorities.



(b) Target: gender minorities.



(c) Target: religious minorities.

Figure 13: Overall, an extremely low percentage of speeches target minority groups. Therefore, I will ignore this analysis. (TODO explain better.. within error yada yada).

Conclusions

Italian prime minister seem to be overall a bit aggressive and negative, and they speak more often then not about the opposition. Almost no traces of hate speech and attacks against minority groups are found in the data.

This does not seem to change over time, failing to explain the VDEM indices drop in 2021.